Band-1 Hadamard Order 2276 via Miyamoto — Paley-I H_284 → Double H_568 + Paley Conference C_570

H_284/H_568/H_2276 PASS; Paley-II H_568 path REJECTED; SPK gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_2228_Miyamoto`; `2026-09-14_Band1_Hadamard_next_gap_after_2036_`
Lab. `/workspace/experiments/band1/hadamard_2276_miyamoto/`.
Cutoff. 2026-09-14 (~evening ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 284, 568, and 2276 only** (verified max $|HH^\top - nI| = 0$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2276 is a **Miyamoto bonus** (not an SPK fill). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed H_{508}, H_{2004}, H_{2036}, H_{528}, H_{2116}, H_{540}, H_{2164}, H_{556}, and H_{2228}. This increment closes the last named Miyamoto-eligible target from the post-2036 triage list: **order 2276** ($2276 = 4 \cdot 569$, 569 prime $\equiv 1 \pmod 4$). Direct Paley-II for H_{568} **fails** ($568 = 2(q+1)$ forces $q = 283 \equiv 3 \pmod 4$). On box: H_{284} via Paley type I ($q = 283$), H_{568} via Sylvester doubling, symmetric Paley conference C_{570} over prime 569, and $H_{2276} = \text{Miyamoto}(H_{568}, C_{570})$. All verify $\max |HH^\top - nI| = 0$. Unit tests **19/19 PASS**. SPK next gap remains **2052**; $2276 \notin$ SPK closure (Miyamoto bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2276; Miyamoto; Paley type I; Sylvester doubling; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{284} constructed and verified (Paley type I, $q = 283$). 2. H_{568} constructed and verified (Sylvester double of H_{284}). Paley-II path **rejected**. 3. C_{570} constructed and verified (Paley conference, $q = 569$): $CC^\top = 569I$, zero diagonal. 4. H_{2276} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK next constructive gap remains **2052** (OPEN). Order 2276 does **not** enter SPK.

2. Arithmetic and eligibility

Quantity	Value
Target order	$2276 = 4 \cdot 569$
$q = 569$	prime, $569 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{568}
Needed conference	order $570 = q + 1$
H_{568} path (used)	Paley I $q = 283 \rightarrow H_{284}$; Sylvester double $\rightarrow H_{568}$
H_{568} Paley-II path	**REJECTED**: $q' = 283 \equiv 3 \pmod 4$ (needs $\equiv 1$)
Conference path	Symmetric Paley over prime 569

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{284}	Paley type I, $q = 283$	0	0.0496	**PASS**
H_{568}	Sylvester double of H_{284}	0	0.0918	**PASS**
C_{570}	Paley conference, $q = 569$	0	0.2554	**PASS**
H_{2276}	Miyamoto(H_{568}, C_{570})	0	5.2036	**PASS**

Total wall (incl. unit tests + SPK refresh) $\approx$ 5.8734 s. Compressed ± 1 matrices saved as `H284_pm1.npz`, `H568_pm1.npz`, `H2276_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\cup$ verified H_{284} / H_{568} (Paley-I + double seeds) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK.**

Quantity	Value
Known count	879
2276 in SPK	**False** (Miyamoto bonus)
568 in SPK	True (Paley-I + double seed)
2228 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172…

5. Unit tests (19/19)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`paley_II_q5_H12`	PASS
`paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS
`paley_I_q7_H8_double_H16`	PASS
`q569_is_prime`	PASS
`q569_equiv_1_mod_4`	PASS
`n2276_equals_4_times_569`	PASS
`q283_is_prime`	PASS
`q283_equiv_3_mod_4`	PASS
`H284_order_formula`	PASS

Test	Result
`H568_double_formula`	PASS
`C570_order_formula`	PASS
`paley_II_H568_path_rejected`	PASS
`reject_all_ones_H8`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2276_arith`	PASS
`conference_570_shape_expected`	PASS

6. Non-claims

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE.
- Not a fill of standard gap **2052**.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.
- Order 2276 is a **Miyamoto bonus** outside SPK (do not fold Miyamoto into SPK closure).

7. Next optional

Method-attack standard gap **2052** (SPK still open); OR Track B privacy / Object 1 meter await; OR other Band-1 scoreboard refresh. Post-2036 Miyamoto-eligible list **2116/2164/2228/2276** now all constructed on box.

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2276_miyamoto
python3 verify_hadamard_2276.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H284_pm1.npz`, `H568_pm1.npz`, `H2276_pm1.npz`.